

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-5850
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture flow for this project:

1. The employee data is created in Google Sheets and downloaded as Sample Spreadsheet.xlsx.
2. The file is uploaded to ServiceNow through System Import Sets > Load Data and stored in the Import Set (staging) table Employee Import (u_employee_import).
3. The Transform Map "Sample Spreadsheet Import" maps the staged fields to the target table Employee Test (u_employee_test).
4. Coalesce on Employee ID decides whether each record is inserted, updated or ignored.
5. Transform History and the Import Log are used to verify the result of every load.
6. Reports (pie, bar, list) are built on the target table and shown on the Employee Analytics Dashboard.

Example: Import Data using Transform Maps (Spreadsheet)

Technical Architecture - Import Data using Transform Maps (Spreadsheet)

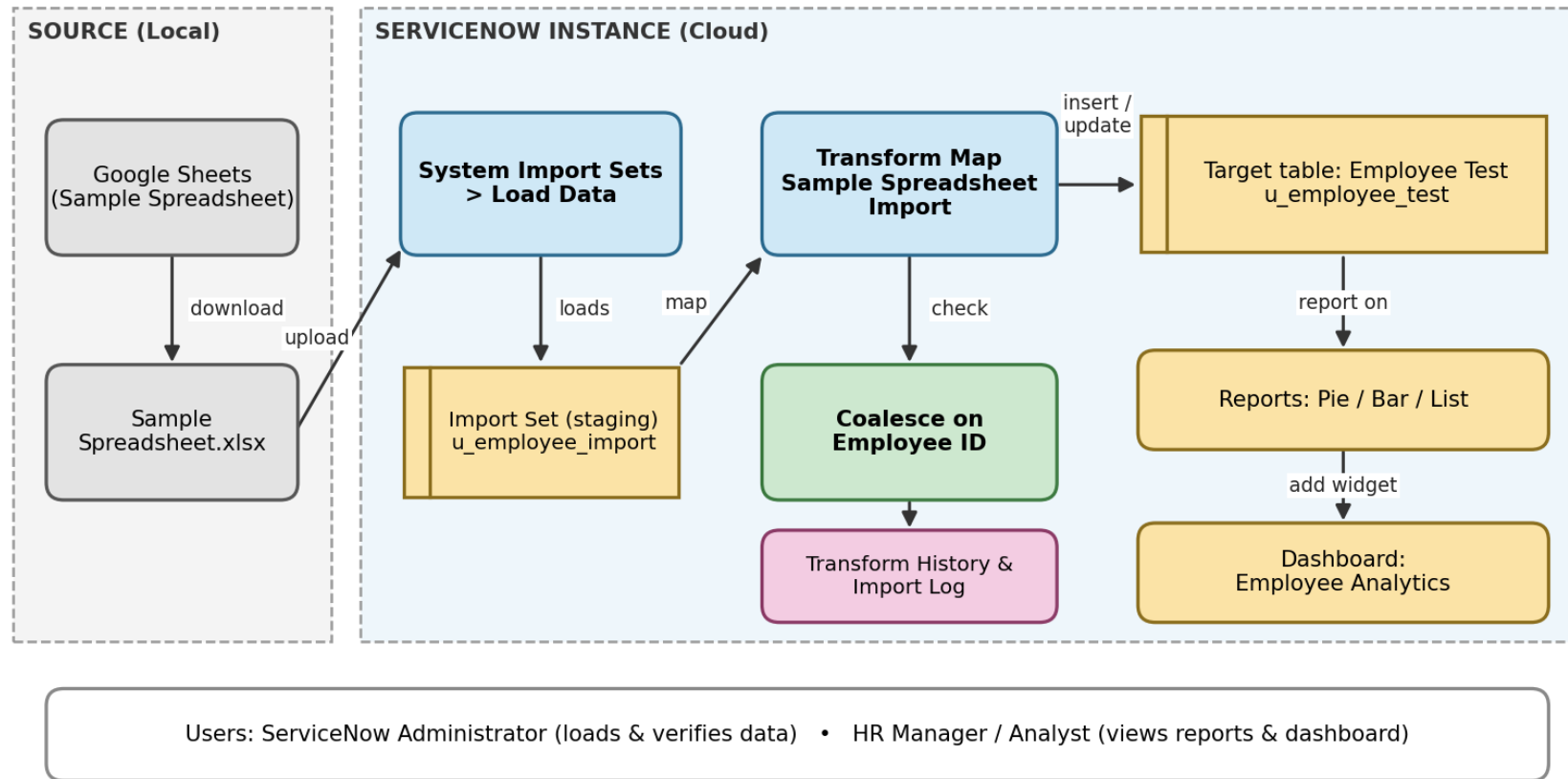


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application: ServiceNow web UI (Application Navigator, forms, list views, Reports and Dashboards) and Google Sheets for the source data	ServiceNow web UI (browser), Google Sheets
2.	Application Logic-1 (Data Import)	Loads the spreadsheet rows into a staging table before any change is made to real data	ServiceNow System Import Sets – Load Data, Import Set table u_employee_import
3.	Application Logic-2 (Transformation)	Maps source fields to target fields and creates the records in the target table	ServiceNow Transform Map “Sample Spreadsheet Import”, Auto Map Matching Fields / Mapping Assist
4.	Application Logic-3 (Duplicate Check)	Prevents duplicate employees by updating a record when the Employee ID already exists	Coalesce on Employee ID (field map)
5.	Database	Custom target table with five String fields: Employee ID, Employee Name, Email, Department, Location	ServiceNow platform database – table u_employee_test
6.	Cloud Database	Database service on the cloud: all tables are hosted by the ServiceNow instance	ServiceNow cloud instance
7.	File Storage	Spreadsheet file used as the data source; it is uploaded once per load	Sample Spreadsheet.xlsx (exported from Google Sheets)
8.	External API-1	Not used – data is brought in by file upload	Not applicable
9.	External API-2	Not used	Not applicable
10.	Machine Learning Model	Not used in this project	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on Local System / Cloud Local Server Configuration: any computer with a web browser Cloud Server Configuration: ServiceNow cloud instance (Global application scope)	Cloud – ServiceNow platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is needed; the solution uses ServiceNow's built-in, low-code modules	ServiceNow out-of-the-box Import Sets, Transform Maps, Reports and Dashboards
2.	Security Implementations	Role-based access, ACLs (pa_dashboards create ACL with Admin overrides) and dashboard sharing restricted to selected users, groups and roles	ServiceNow Roles, ACLs, Share settings
3.	Scalable Architecture	Staging table, Transform Map and target table are separate layers (browser UI, ServiceNow application, database), so the same map can be reused for larger files and new fields	3-tier ServiceNow architecture; Import Set + Transform Map
4.	Availability	Hosted on the cloud instance and accessed through a browser, so the data, reports and dashboard are available without any local setup	ServiceNow cloud
5.	Performance	Bulk batch transform instead of manual entry: 15 rows in about 1.4 s and 4 rows in about 0.04 s, with the counts recorded in Transform History	Import Set batch transform, Transform History

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>